

WARCOP RANGE, United Kingdom

WMO: 032260

Lat: 54.5725N

Lon: 2.4131W

Elev: 227

StdP: 98.63

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	-3.2	-6.5	2.2	-2.4	-5.1	2.5	-1.1	18.5	3.6	16.1	3.8	1.3	140	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	23.2	17.8	21.2	16.6	19.5	15.6	18.6	21.8	17.3	20.3	16.2	18.7	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.8	19.7	16.1	11.8	18.6	15.1	11.0	17.5	53.5	22.0	49.4	20.3	46.3	18.7	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	12.8	11.0	-7.3	26.0	1.6	1.9	-8.4	27.3	-9.4	28.4	-10.2	29.5	-11.4	30.9		
(5)			-7.4	21.5	1.5	2.6	-8.5	23.4	-9.4	24.9	-10.3	26.4	-11.4	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	3.5	3.7	5.0	7.2	10.0	12.6	14.6	14.2	12.1	9.1	6.0	3.6		
	DBStd	4.93	2.97	2.97	2.86	2.85	2.97	2.52	2.45	2.39	2.42	2.92	3.15	3.47		
	HDD10.0	1048	201	177	157	93	37	4	0	1	7	50	123	197		
	HDD18.3	3601	459	411	414	336	259	173	120	131	187	286	371	455		
	CDD10.0	498	0	1	1	8	36	82	143	131	71	23	3	0		
	CDD18.3	8	0	0	0	0	0	1	4	3	1	0	0	0		
	CDH23.3	47	0	0	0	0	1	6	23	17	1	0	0	0		
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0		
(14) Wind	WSAvg	4.5	5.1	5.0	5.0	5.0	5.0	4.2	3.7	3.7	4.0	4.6	4.5	4.5		
(15) Precipitation	PrecAvg	1367	161	122	97	81	81	79	90	102	103	128	148	173		
	PrecMax	1699	255	296	183	150	149	178	201	245	190	250	368	450		
	PrecMin	997	33	22	29	37	22	25	49	22	26	47	69	53		
	PrecStd	179	61	75	34	31	35	39	34	45	39	50	67	85		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.9	15.1	18.9	22.4	23.9	25.8	25.7	21.5	17.5	13.7	11.4		
		MCWB	9.5	9.1	9.2	12.4	17.1	18.0	19.1	18.6	17.9	14.5	12.3	10.4		
	2%	DB	9.6	10.0	12.1	15.7	19.8	21.1	23.4	22.4	19.0	15.5	12.5	10.4		
		MCWB	8.5	8.4	8.7	10.8	15.3	16.1	17.9	17.8	15.9	13.4	11.3	9.4		
	5%	DB	8.7	8.9	10.3	13.4	17.4	19.2	21.4	19.9	17.3	14.4	11.4	9.5		
		MCWB	7.6	7.7	8.1	9.6	13.3	15.0	16.8	16.2	14.5	12.7	10.2	8.4		
	10%	DB	7.8	7.9	9.1	11.8	15.2	17.4	19.4	18.3	16.1	13.4	10.4	8.5		
		MCWB	6.8	6.8	7.4	8.8	11.7	13.7	15.8	15.4	13.7	11.8	9.2	7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.0	11.0	13.2	19.1	18.6	21.5	20.7	18.6	15.2	12.8	10.7		
		MCDB	10.7	11.0	13.6	17.7	21.0	22.0	23.2	21.9	20.5	16.5	13.5	11.2		
	2%	WB	8.7	8.8	9.4	11.4	15.7	16.7	18.7	18.0	16.6	14.0	11.6	9.7		
		MCDB	9.4	9.7	11.1	14.4	19.1	20.4	22.4	20.8	18.3	15.0	12.4	10.3		
	5%	WB	7.9	7.9	8.5	10.2	13.7	15.3	17.3	16.9	15.2	13.0	10.4	8.6		
		MCDB	8.5	8.8	9.9	12.6	16.7	18.4	20.6	19.3	16.6	14.2	11.2	9.4		
	10%	WB	6.9	7.0	7.6	9.2	12.1	14.2	16.1	15.9	14.0	12.0	9.4	7.6		
		MCDB	7.7	7.8	9.0	11.4	14.7	16.8	18.7	17.9	15.6	13.2	10.3	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.9	6.0	7.1	7.6	7.3	7.4	6.9	6.6	5.7	4.8	4.7		
		MCDBR	4.9	5.5	7.7	9.6	11.0	10.5	10.9	9.5	8.0	6.1	4.9	5.2		
	5% WB	MCWBR	5.0	4.8	5.4	6.4	7.5	6.8	6.8	6.0	5.6	4.9	4.7	5.2		
		MCWBR	4.9	5.3	6.8	8.3	9.9	9.5	9.4	8.0	7.1	5.8	4.8	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.331	0.361	0.383	0.385	0.384	0.384	0.372	0.355	0.339	0.314	0.302		
		taud	2.262	2.337	2.276	2.241	2.281	2.313	2.316	2.363	2.414	2.413	2.336	2.242		
	Ebn at Noon		610	751	807	839	859	863	854	843	811	733	617	541		
		Edh at Noon	59	77	102	121	123	120	118	106	89	71	55	49		
(44) All-Sky Solar Radiation	RadAvg		0.56	1.22	2.33	3.73	4.84	4.88	4.71	3.80	2.70	1.45	0.71	0.42		
	RadStd		0.07	0.14	0.22	0.37	0.36	0.47	0.48	0.28	0.23	0.17	0.07	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page